Problem Pack: B. Square Roots and Square Applications

Question 6

Problem: 6. Complete each inequality with the two consecutive whole numbers that the square root lies between. a) [ ] < √40 < [ ] b) [ ] < √55 < [ ] c) [ ] < √110 < [ ]

Student Work:

Feedback

Question 7

Problem: 7. Approximate each square root to the nearest tenth. Use nearby benchmarks or nearby tenths to justify your answer. a) √45 b) √83 c) -√72

Student Work:

a)

b)

c)

Feedback

In question 7, you went wrong in part a when you wrote too many decimal places instead of rounding to the nearest tenth. Remember that the tenths place is the first number right after the decimal point, so check which benchmark square root is closest to help you round correctly.

Question 8

Problem: 8. A square garden has an area of 52 m^2. a) Write the exact side length. b) State whether the side length is rational or irrational. c) Approximate the side length to the nearest tenth. d) State the two consecutive integers it lies between

Student Work:

a = s^2 √52 = √s^2 s = √52 a) side length = √52 b) irrational c) 7.2 d) Between 7 and 8

Feedback

Question 9

Problem: 9. Find the area of a square with each side length. a) 5 cm b) 1.2 m c) 3/4 m d) √6 cm

Student Work:

a) **25 cm**

b) **1.44 cm**

c) 9/16 m

d) 6 cm

Feedback

Question 10

Problem: 10. Move between the measures of each square. a) A square has a perimeter of 28 cm. Find its side length and its area. b) A square has an area of 45 cm^2. Find its exact side length and its exact perimeter.

Student Work:

a) P = 4(s) 28/4 = 4(s)/4 s = 7 cm a = s^2 = 7^2 = 49 cm

b) a = s^2 √45 = √s^2 s = √45 P = 4(s) = 4(√45)

Feedback

Unsure About Shown Work

Question 11

Problem: 11. The figure is made from two squares. Use each area to determine the side lengths, then find the perimeter of the outer edge only. a) Find both side lengths. b) Identify the shared side that is not counted on the outside. c) Find the perimeter of the entire shape.

Student Work:

14 14 cm 6 8 8 8 22 a) √196 = 14 cm √64 = 8 cm b) 8 c) 14 + 14 + 22 + 8 + 8 + 6 = 72 cm

Feedback